DYNAP-SE 1 Issues and observations

Usage of Neuron 0 of Core 0 may cause serious unexpected network dynamics and core shutdowns

Reason: All neurons' CAMs are defaulted to a (0, 0) address on startup, so every spike from any (0,0) neuron from any of the chips or the spikegen is routed to ALL unused synapses.

Solution: Do not allocate neuron (0, 0) of any chip or the spikegen into any of the populations. Assume you have 1023 neurons per chip.

Spikegen population - do not use the same neuron IDs as the on-chip population to avoid self-excitation (and self-inhibition)

Reason: Postsynaptic neuron's CAMs are only programmed with Chip ID and Core ID. If the on-chip population is connected in such a way, the spikes of these neurons themselves will be routed back to their input (unwanted self-excitatory\self-inhibitory connections)

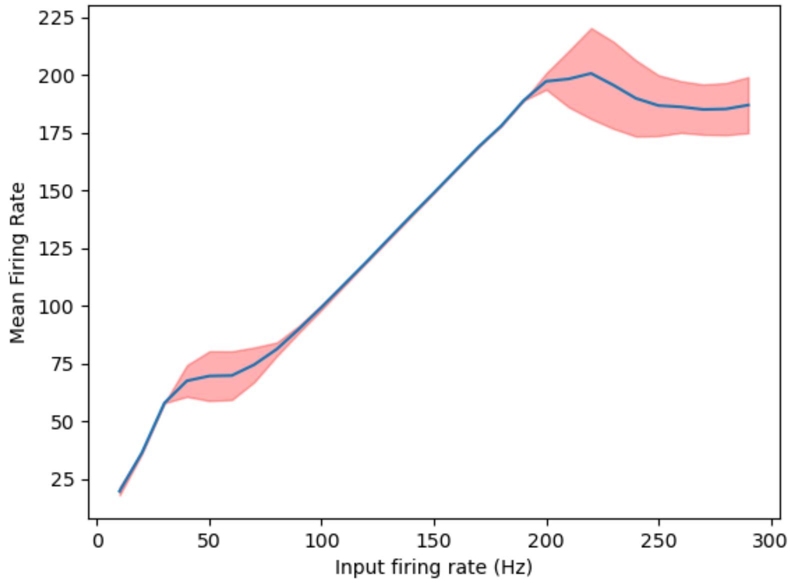
Solution: Shift the input (spikegen) population to have non-overlapping neuron-IDs.

Low leakage ('tau') currents lead to self-activation of synapses.

Reason: One of the properties of analog circuits. With no leak the charge eventually builds up (or depletes, for synapses) in the respective capacitor.

Solution: Always check if the leakages are set to high when neurons are unused.

FF curve dropouts - observed firing rates of neurons drop down with increasing input rates when sharing the input source and measured together. One by one the curves are smooth.



Reason: Not confirmed. Assumed interference of the routed spikes, either at input-to-chip stage or chip-to-monitor (as the timestamps have resolution of microseconds and have to be different).

Solution: (If this happens) measure response curves for one neuron at a time. At low enough rates it should be fine. But be aware that some of the spikes from synchronously firing neurons might be dropped. [REQUIRES INVESTIGATION]

Doubled FPGA input for preloaded sequences. The sequence plays twice if the repeat mode is set to False.

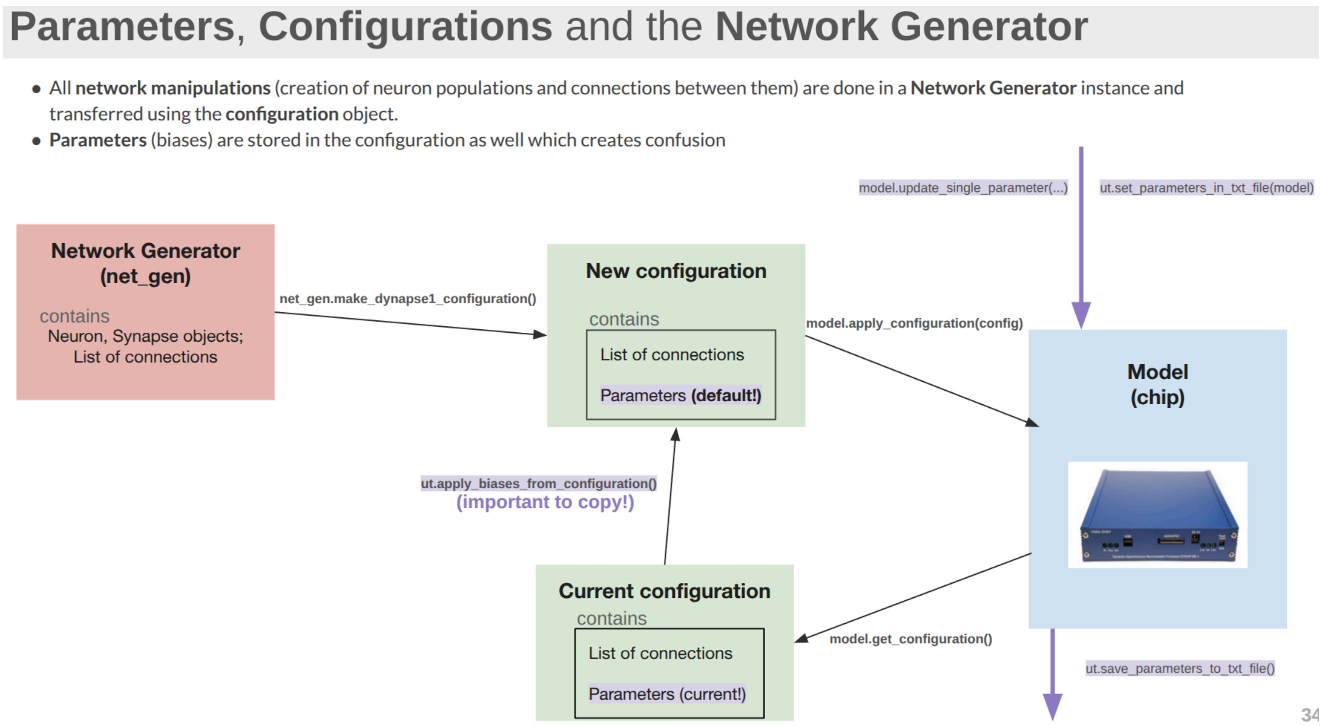
Reason: Unclear. Assumed to be a bug in the FPGA code.

Solution: Since the spiketrain length is known, manually stop the spikegen.

New configuration from Network Generator overrides previously set biases

Reason: Network Generator is an object with no connection to the Model. It populates a blank Configuration with the connectivity matrix, leaving the analog parameters "default".

Solution: Before applying the new Configuration, copy the parameters from the freshly retrieved one from the model (a function in [dynapse1utils.py]() is introduced to do exactly that)



Bad GLXFW Config error when starting the Samna GUI (live spiking activity)

Reason: Wrong configuration of OpenGL in the system.

Solution: 1. Make sure you followed the Samna installation step installing several libraries:

- ubuntu : `apt install mesa-common-dev libxrandr-dev libxinerama-dev libxcursor-dev libxi-dev libglu1-mesa-dev libxcb-dri2-0-dev libxcb-dri3-dev libxcb-present-dev libxcb-sync-dev libx11-xcb-dev libxcb-glx0-dev`
- centos : `yum install  libXrandr libXi  libXinerama libXcursor libGL mesa-libGL-devel mesa-libGLU`

2. For dual GPU laptops use NVIDIA Prime to switch to NVidia GPU. Also try putting system into performance mode.

LEFT CHIPS CANNOT RECIEVE CONNECTIONS FROM THE RIGHT ONES??

Reason: Some jumper positions?

Solution: ?????